

# Olist dataset

## Problem Statement

E-commerce businesses generate massive amounts of transactional data every day, but this data is only useful if it can be properly organized, connected, and queried to answer real business questions. Olist, a Brazilian e-commerce platform, provides a rich dataset of over **100,000 orders** placed between **2016 and 2018**, spanning customers, sellers, products, payments, and reviews across multiple relational tables.

The goal of this project is to design and analyze a relational database using the Olist dataset to understand how customers, orders, payments, and reviews relate to one another, and to answer business questions such as which product categories perform best, how delivery delays affect customer satisfaction, and which sellers consistently deliver good service.

Along the way, the project also involves cleaning the data, resolving duplicate and inconsistent records, and ensuring referential integrity across tables

The aim is to have a well-structured database along with a set of SQL queries that can extract meaningful insights from the data, essentially turning raw transactional records into information that will drive business decisions.

## Columns :

### Olist customer dataset

- Customer\_id (PK) - For every new order a new customer\_id is created like A unique key **per order**.
- Customer\_unique\_id - This column specifies a unique customer ,an *actual* unique identifier of a person, used to track repeat customers across multiple orders .
- Customer\_zip\_code\_prefix - The first five digits of a Brazilian postal code , didnt reveal with postal code , to protect users exact informations.
- Customer\_city & customer\_state - customers location .

### Olist orders dataset

- Order\_id (PK) - Unique Identifier for each order.
- Customer\_id (FK) - Use to link orders to the customer table
- Order\_status - current status of the order (delivered ,shipped ,invoiced ,processing ,approved(payment approved),created ,canceled ,unavailable )
- Order\_purchase\_timestamp - Date and time when the customer placed the order.

- Order\_approved\_at - Time when the payment was approved.
- Order\_delivered\_carrier\_date - Date when the seller handed the package to the logistics carrier.
- Order\_delivered\_customer\_date - Actual delivery date to the customer.
- Order\_estimated\_delivery\_date - The date Olist promised the customer.

Note :

The difference between purchase and approval indicates how long payment approval took.

### **Olist order\_items dataset**

This table tells you what products were purchased, who sold them, and how much each item cost.

- Order\_id (FK) - Indicates which order the item belongs to.
- Order\_item\_id - Represents the item number within an order. It uniquely identifies each item within the order.
- Product\_id - ID of the purchased product.
- Seller\_id - The seller who has sold this product .
- Shipping\_limit\_date - The latest date by which the seller should hand the item over to the shipping carrier.
- Price - price of an item
- Freight\_value - Shipping charge allocated to that specific item.

Note : A single order can contain items from **multiple sellers**.

### **Olist order\_payments dataset**

- Order\_id (FK) - shows which order the payment belongs to.
- Payment\_sequential - The sequence number of the payment for a particular order. Counts **payment records** within an order
- payment\_type - Payment method used. (credit card, debit card, voucher, boleto, non-defined)
- Payment\_installments - Number of installments chosen for the payment.
- Payment\_value - Amount paid in that payment transaction.

## Olist products dataset

It contains descriptive information about every product sold on the Olist marketplace.

- Product\_id - Unique identifier for each product.
- product\_category\_name - Product category in Portuguese.
- Product\_name\_length - Number of characters in the product name.
- product\_description\_length - Number of characters in the product description.
- product\_photos\_qty - Number of photos available for the product listing.
- product\_weight\_g - Weight of the product in **grams**.
- product\_length\_cm - Length of the package in centimeters.
- product\_height\_cm - Height of the package in centimeters.
- product\_width\_cm - Width of the package in centimeters.

## Olist seller dataset :

- seller\_id - Unique identifier for each seller
- seller\_zip\_code\_prefix - The first **5 digits** of the seller's Brazilian postal code.
- seller\_city - City where the seller is located
- seller\_state - Brazilian state abbreviation.

## Olist product category name translation dataset :

- product\_category\_name - This is the category name stored in the **products** table.
- product\_category\_name\_english - English translation of the category.

## Olist order reviews dataset :

- review\_id(PK) - Identifier of the review.
- order\_id (FK) - Indicates which order the review belongs to.
- review\_score - Customer rating from **1 to 5**.
- review\_comment\_title - title written by the customer.
- review\_comment\_message - customer feedback.
- review\_creation\_date - Date when the customer submitted the review.
- review\_answer\_timestamp - Timestamp when the reviewed by Olist.

### **Olist geolocation dataset :**

contains the **geographical coordinates (latitude and longitude)** for Brazilian ZIP code prefixes (CEP prefixes). It allows you to map customers and sellers to approximate locations on a map and perform geographic analysis.

- geolocation\_zip\_code\_prefix - The first **5 digits** of a Brazilian postal code (CEP).
- geolocation\_lat - Latitude coordinate.
- geolocation\_lng - Longitude coordinate.
- geolocation\_city - City corresponding to the ZIP code prefix.
- geolocation\_state - Brazilian state abbreviation.

Note : Create a lookup table with one row per ZIP code prefix before joining.